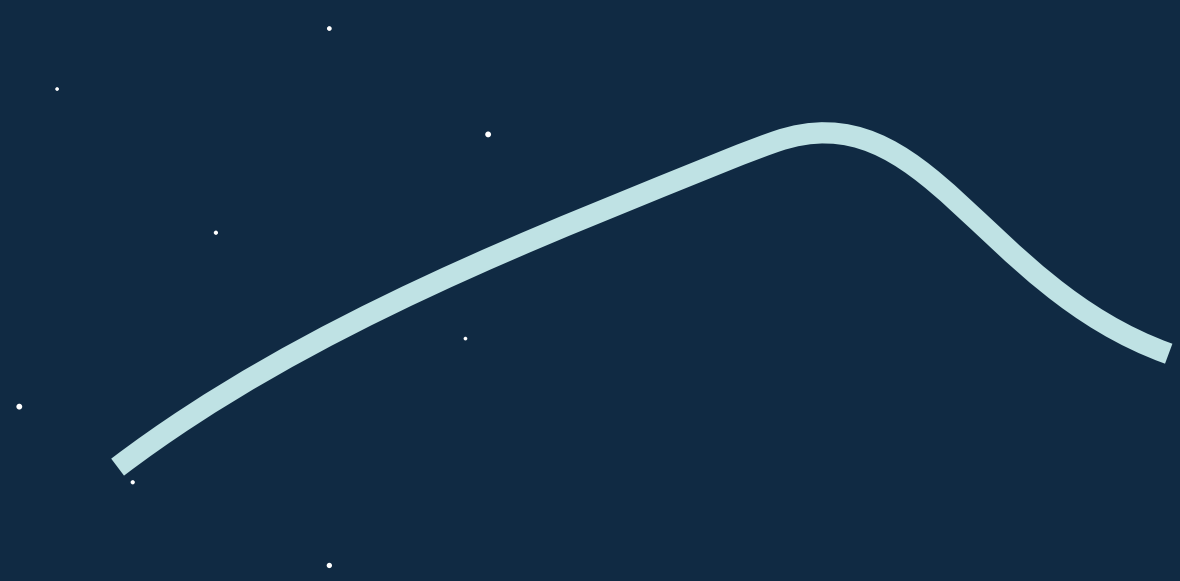
DETAILED PROJECT AGENDA

Topological Asteroseismology

Stage-by-stage execution, validation, and publication roadmap for solar-like stars

A gated operational plan linking scientific questions to tasks, deliverables, verification evidence, decision criteria, reproducibility controls, and fallback paths.

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Planning baseline	August 2026 – January 2029
Defense target	May 2027



Document purpose

This agenda operationalizes the companion research proposal, *Topological Asteroseismology of Solar-Like Stars: Persistent Structure in Light Curves, Power Spectra, and Echelle Geometry*. It is a project-control document rather than a repetition of the scientific proposal. Each stage specifies purpose, prerequisites, tasks, deliverables, acceptance tests, exit criteria, dependencies, risks, and scope decisions. Numerical thresholds marked "freeze after pilot" must be fixed before confirmatory analyses and cannot be adjusted after inspecting held-out results.

How to use the agenda

At the beginning of each stage, confirm inputs and freeze the task scope. During execution, maintain an experiment ledger and produce machine-readable evidence for each acceptance test. At the end, hold a gate review with one of four outcomes: *pass*, *pass with documented limitation*, *repeat a bounded subset*, or *stop/reframe*. A stage is not complete merely because code runs; its scientific deliverables must be interpretable, reproducible, and validated.

Three levels of priority

Core (C). Required for the minimum defensible thesis and first astronomy paper.

Publication (P). Required for population-level claims and a strong methods paper, but may continue after the defense.

Extension (E). High-value work that is explicitly outside the defense-critical path unless earlier stages finish ahead of schedule.

The planned May 2027 defense creates a ten-month execution window if Stage 0 begins in August 2026. The defense-critical path therefore ends with a mathematically specified, null-calibrated framework, the complete Rudy benchmark, and a compact independently locked Kepler validation. Exhaustive LEGACY analysis, TESS expansion, extensive MESA/GYRE grids, and all three planned papers belong to the full 30-month publication program.

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Abbreviations and stage notation

Term	Meaning
TDA	Topological data analysis.
PH	Persistent homology.
PSD	Power spectral density.
SC	Kepler short cadence.
KIC / TIC	Kepler Input Catalog / TESS Input Catalog.
LEGACY	Kepler LEGACY sample of 66 well-characterized solar-like oscillators.
Rudy	KIC 10963065; benchmark star, also HD 176693, Kepler-408, TIC 48450369.
Sn	Stage n.
T$n.m$	Task m within Stage n.
D$n.m$	Deliverable m within Stage n.
Gn	Decision gate following Stage n.
AR	Acceptance requirement: a testable condition for a deliverable.
Critical path	Minimum sequence that must finish before the May 2027 thesis defense.
Holdout	Data not used for method selection, tuning, or threshold definition.

1. Program architecture and governance

1.1. Scientific product structure

The program is designed around four nested claims. Later claims are attempted only after earlier ones have adequate evidence.

Claim I. Selected topological constructions have a mathematically identifiable relationship to controlled oscillatory or spectral structure.

Claim II. Their estimators and null distributions are calibrated under realistic stochastic excitation, noise, sampling, and background conditions.

Claim III. They recover or certify solar-like asteroseismic information in Rudy and generalize to independently selected Kepler stars.

Claim IV. They add practical information beyond classical statistics or provide a rigorous account of redundancy and failure domains, including Kepler-to-TESS transfer.

1.2. Stage dependency map

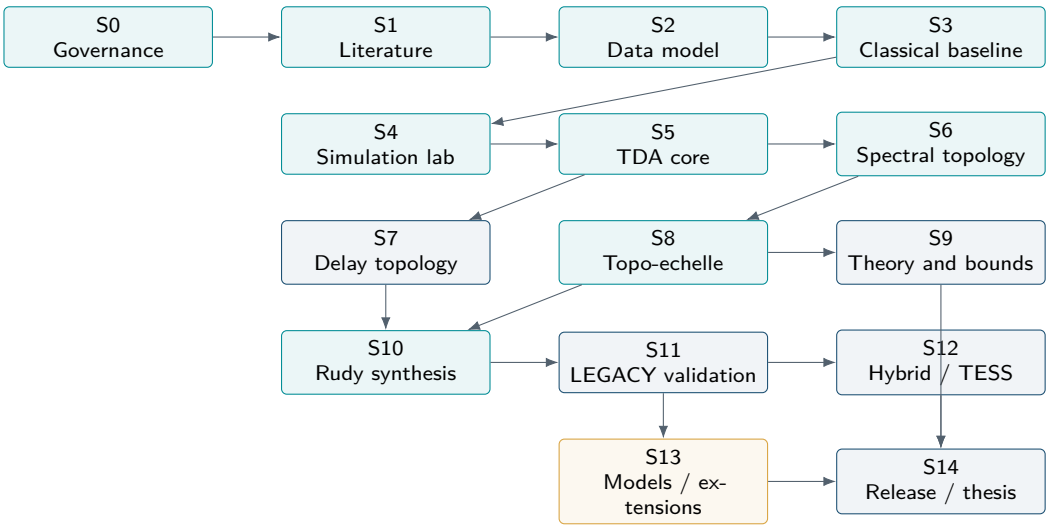


Figure 1.1.: Primary stage dependencies. Teal stages are on the scientific core; blue stages establish population and transfer claims; gold marks a bounded extension.

1.3. Project-wide rules

1. **No topological tuning on the final holdout.** Representation choice, diagram summaries, thresholds, and hyperparameters are frozen using analytic cases, simulations, and development stars.

2. **Every topology statistic receives a simpler control.** Controls include peak prominence, autocorrelation, ridge linearity, Fourier summaries, or a non-topological image statistic matched to the proposed information.

3. **Every claimed detection receives a null.** Background-only, randomized comb, mission-window, phase-randomized, and IAAFT nulls are selected according to the claim.
4. **Results remain localized.** Persistence points must retain links to time, frequency, radial order, ridge identity, or pixels; anonymous feature vectors alone are insufficient.
5. **Uncertainty includes analysis choices.** Preprocessing and background alternatives are treated as registered branches, not informally selected after seeing the answer.
6. **Negative results are archived.** Failed representations, unstable statistics, and counterexamples remain in the experiment ledger and inform scope decisions.
7. **Writing overlaps analysis.** Each stage ends with manuscript-ready methods, figures, tables, and limitations text.

1.4. Definition of done

A stage is done only when: (i) all core deliverables exist; (ii) acceptance requirements pass or exceptions are signed off; (iii) inputs, configurations, software versions, and random seeds are recorded; (iv) figures regenerate from archived tables; (v) limitations and negative results are documented; and (vi) the next stage can start without hidden manual steps.

2. Stage 0 - Project governance and reproducibility foundation

Stage objective

Transform the proposal into a controlled, auditable research program with frozen terminology, ownership, repository conventions, and a preregistered validation logic.

Priority: Core. **Indicative duration:** 2 weeks. **Depends on:** approved proposal.

2.1. Tasks

ID	Task	Required work and evidence	Output
T0.1	Freeze research questions	Translate each scientific objective into a falsifiable hypothesis, estimand, null hypothesis, baseline, unit of analysis, and minimum publishable conclusion.	Hypothesis registry
T0.2	Define terminology	Fix meanings of persistence, ridge, coherence, topological echelle, detection, calibration, robustness, complementarity, and failure.	Data/term dictionary
T0.3	Establish repository	Create package, configurations, tests, workflows, notebooks, manifests, reports, figure recipes, and environment lock.	Tested skeleton
T0.4	Establish governance	Define branch/review policy, coding standard, issue labels, experiment IDs, authorship discussion schedule, and decision authority.	Governance file
T0.5	Preregister validation	Separate exploration, method development, confirmation, and final holdout; define one-time holdout access and deviation reporting.	Analysis protocol
T0.6	Plan data management	Define raw/derived/archive tiers, checksums, MAST identifiers, naming, retention, backup, licensing, and DOI targets.	DMP
T0.7	Build risk register	Assign probability, impact, early warning, mitigation owner, fallback, and review date to scientific and program risks.	Risk register

ID	Task	Required work and evidence	Output
T0.8	Inventory resources	Verify storage, CPU allocation, software licenses, archive access, and potential MESA/GYRE capability; benchmark a small persistence run.	Resource note

2.2. Deliverables and acceptance

- **D0.1 - Project charter:** scope, claims, priorities, responsibilities, meeting rhythm, and change-control rule.
- **D0.2 - Reproducibility specification:** environment lock, seed policy, manifest schema, configuration hashing, test tiers, and archive plan.
- **D0.3 - Preregistered analysis protocol:** partitions, primary metrics, exclusions, ablations, nulls, and holdout policy.
- **D0.4 - Running repository skeleton:** installation, lint/type/test commands, a smoke workflow, and continuous integration.

Stage gate

G0 pass condition. A collaborator can clone the repository, create the environment, run the smoke workflow, and identify every planned dataset and primary hypothesis without oral instructions. All unresolved choices that could change a headline conclusion are either frozen or explicitly scheduled for a pilot-stage decision.

Risks and fallback

If governance becomes documentation-heavy, retain only artifacts that control scientific flexibility or reproducibility. If compute access is uncertain, use small simulations and one-dimensional persistence until profiling supports larger experiments.

3. Stage 1 - Literature evidence map and benchmark specification

Stage objective

Establish the exact scientific gap, prevent reinvention, and convert the literature into quantitative benchmark targets for solar-like asteroseismology and TDA.

Priority: Core. **Duration:** 2–3 weeks, then continuous updates. **Depends on:** S0.

3.1. Tasks

ID	Task	Required work and evidence	Output
T1.1	Solar-like oscillation map	Review excitation, damping, mode linewidths, asymptotic combs, granulation, mode visibility, and observational limitations. Record definitions and assumptions.	Physics matrix
T1.2	Classical pipeline map	Compare Lightkurve, pySYD, ACF, PSPS, background fitting, echelle construction, and peak-bagging products; record algorithms and known failure modes.	Baseline matrix
T1.3	Rudy evidence table	Reconcile identifiers, observations, published $\nu_{\max}$, $\Delta\nu$, mode frequencies, stellar parameters, and uncertainty conventions.	Benchmark sheet
T1.4	LEGACY sample map	Identify authoritative mode-frequency and stellar-property tables, selection function, evolutionary range, and data access.	Population sheet
T1.5	Mathematical TDA map	Review stability, one-dimensional persistence, sliding-window embeddings, periodic/cubical complexes, zigzag persistence, vineyards, and statistical inference on diagrams.	Theorem matrix
T1.6	Astronomy TDA map	Document adjacent uses in time series, light curves, spectra, morphology, and cosmology; distinguish direct precedent from analogy.	Gap map
T1.7	Novelty audit	Search explicitly for topological echelle diagrams, PH-based seismic estimators, and topology of solar-like oscillations; record queries and dates.	Novelty memo

ID	Task	Required work and evidence	Output
T1.8	Reference validation	Verify metadata, DOI, equations, and claimed support for every foundational citation; reject secondary citations where primary sources exist.	Clean BibTeX

3.2. Deliverables and acceptance

- **D1.1 - Structured literature matrix** with question, method, dataset, sample size, result, limitation, and relevance fields.
- **D1.2 - Research-gap memorandum** separating established results, plausible extensions, and genuinely novel claims.
- **D1.3 - Benchmark target table** for Rudy and the LEGACY sample, including units, uncertainty type, source, and reconciliation notes.
- **D1.4 - Updated bibliography** with verified primary sources and a quarterly update protocol.

Stage gate

G1 pass condition. The topological echelle and hybrid-validation questions remain non-duplicative after a reproducible search; each benchmark number has a traceable source; and at least one classical baseline is available for every planned topological claim. If direct precedent is found, redefine novelty around validation, stability, or mission transfer rather than priority.

4. Stage 2 - Data acquisition, provenance, and canonical data model

Stage objective

Build a mission-aware, versioned data layer for Rudy, simulations, Kepler development/holdout stars, and TESS transfer experiments.

Priority: Core. **Duration:** 3 weeks. **Depends on:** S0–S1.

4.1. Tasks

ID	Task	Required work and evidence	Output
T2.1	Resolve identifiers	Crossmatch KIC 10963065, HD 176693, Kepler-408, TIC 48450369, coordinates, Gaia identifiers, and published aliases.	Alias record
T2.2	Inventory Kepler products	Record quarters, cadence, author, flux type, exposure, quality masks, time system, file ID, checksum, and coverage for Rudy and sample stars.	Kepler manifest
T2.3	Inventory TESS products	Record sectors, cadence, SPOC/QLP availability where relevant, contamination, crowding, quality, and spectral-window implications.	TESS manifest
T2.4	Define canonical light curve	Specify time, flux, flux uncertainty, quality, cadence, mission, segment, aperture/product provenance, mask reason, and preprocessing state.	Table schema
T2.5	Define spectral products	Specify frequency units, one/two-sided normalization, PSD units, Nyquist handling, window function, background components, and provenance links.	Spectrum schema
T2.6	Acquire reference data	Store or reference published mode frequencies, ratios, $\nu_{\max}$, $\Delta\nu$, stellar parameters, and uncertainties with source/version metadata.	Reference tables
T2.7	Run integrity checks	Detect duplicates, missing cadence, time-order errors, NaNs, nonphysical uncertainty, segment overlaps, unit inconsistencies, and checksum changes.	QC report

ID	Task	Required work and evidence	Output
T2.8	Freeze population partitions	Create development, internal validation, and final holdout partitions grouped by star and stratified by key stellar/observational properties.	Signed split file

4.2. Required tests

Schema tests verify units, monotonic time within segments, unique product keys, nonnegative uncertainties, valid mission/cadence combinations, and round-trip serialization. Provenance tests confirm that every derived row points to raw product IDs, configuration hashes, and software version. The partition test rejects any star appearing in more than one role.

4.3. Deliverables and gate

- **D2.1** Rudy Kepler/TESS manifest and alias table.
- **D2.2** Canonical light-curve, spectrum, mode-frequency, and stellar-reference schemas.
- **D2.3** Quality-control report and exclusion ledger.
- **D2.4** Frozen star-grouped development/validation/holdout split.

Stage gate

G2 pass condition. Every planned input has stable identity and provenance; schemas pass integrity tests; no cross-partition star leakage exists; and the exact Rudy products used by the Lightkurve reproduction are recorded. Ambiguous or unavailable TESS products do not block the gate, but must be marked as optional transfer inputs.

5. Stage 3 - Classical asteroseismic baseline and observation model

Stage objective

Establish a trusted non-topological baseline for Rudy and quantify how sampling, background treatment, and preprocessing affect conventional asteroseismic observables.

Priority: Core. **Duration:** 4 weeks. **Depends on:** S2.

5.1. Tasks

ID	Task	Required work and evidence	Output
T3.1	Reproduce Lightkurve case	Use the documented Rudy quarters and reproduce cleaning, periodogram, $\nu_{\max}$, $\Delta\nu$, mass, radius, and log g with versioned code.	Baseline notebook
T3.2	Run independent pipeline	Apply pySYD or an equivalent independent pipeline; reconcile frequency/PSD conventions and uncertainty outputs.	Pipeline comparison
T3.3	Model background	Fit Harvey-like granulation and white-noise components; compare likelihoods, residuals, and posterior/pipeline uncertainty.	Background models
T3.4	Quantify window function	Compute spectral windows by segment and combination; label aliases and assess quarter/sector stitching choices.	Window atlas
T3.5	Build ordinary echelle	Produce echelle diagrams across trial $\Delta\nu$, standardized orientation, smoothing, order range, and circular edge treatment for later comparison.	Baseline echelle set
T3.6	Compare mode references	Overlay published frequencies and mode degrees; calculate residuals and reconcile radial-order indexing conventions.	Mode audit
T3.7	Register preprocessing branches	Freeze alternative detrending, quality masks, gap treatment, flux products, and PSD normalizations; do not select by topological appearance.	Branch registry

ID	Task	Required work and evidence	Output
T3.8	Estimate classical uncertainty	Bootstrap appropriate segments/realizations, compare pipeline uncertainty and reference disagreement, and separate random from pipeline-choice terms.	Uncertainty table
T3.9	TESS feasibility pilot	Analyze sectors individually; assess envelope detectability and resolution without forcing a combined detection.	Feasibility note

5.2. Reference targets and interpretation

The expected Rudy neighborhood is approximately $\nu_{\max} \sim 2145$–$2160\,\mu$Hz and $\Delta\nu \sim 103.1$–$103.6\,\mu$Hz, but agreement is judged using documented uncertainty and processing choices rather than these numbers as exact answers. Scaling-relation estimates are validation products, not ground truth.

5.3. Deliverables and gate

- **D3.1** One-command Rudy classical workflow and benchmark notebook.
- **D3.2** Baseline table with global seismic parameters, reference values, residuals, and uncertainty components.
- **D3.3** Window/background/preprocessing sensitivity report.
- **D3.4** Conventional echelle and mode-overlay figure suite.
- **D3.5** TESS detectability and combination decision note.

Stage gate

G3 pass condition. Rudy global parameters agree with defensible reference ranges within joint uncertainty, or every discrepancy is traced to a documented convention or modeling choice. PSD units and background normalization pass analytic/injection tests. If not, topology work pauses because an unreliable observation model would contaminate every later result.

6. Stage 4 - Analytic cases and stochastic simulation laboratory

Stage objective

Create known-truth experiments that reveal what each topological construction measures and calibrate false alarms, coverage, aliases, and operating limits before real-data claims.

Priority: Core. **Duration:** 6–8 weeks. **Depends on:** S3 definitions; may overlap late S3.

6.1. Simulation hierarchy

1. **Exact analytic signals:** constant, impulse, single sinusoid, two sinusoids, finite comb, Gaussian/Lorentzian peaks, straight and curved ridges.
2. **Stochastic oscillator spectra:** damped modes with prescribed frequency, height, linewidth, lifetime, degree visibility, rotation where needed, and Harvey-like background.
3. **Mission observation model:** cadence integration, finite duration, gaps, duty cycle, real Kepler/TESS windows, photon noise, and contamination.
4. **Physics-informed spectra:** asymptotic mode patterns or a compact MESA/GYRE-derived grid for structural stress tests.

6.2. Tasks

ID	Task	Required work and evidence	Output
T4.1	Prove exact cases	Derive 1D peak-persistence/prominence equivalence, sinusoidal delay-ellipse geometry, comb folding, and ridge mismatch behavior under explicit assumptions.	Analytic notes/tests
T4.2	Implement simulator	Generate stochastic damped modes plus background in time or frequency domain; expose physical and observational parameters.	Simulator API
T4.3	Validate simulator	Recover target PSD, mode linewidth/height statistics, time-domain variance, cadence effects, and random-realization distributions.	Validation report
T4.4	Define factorial pilot	Vary duration, cadence, S/N, duty cycle, $\nu_{\max}$, $\Delta\nu$, mode lifetime, number of visible orders, background mismatch, and ridge curvature.	Design matrix

ID	Task	Required work and evidence	Output
T4.5	Implement null families	Background-only, randomized comb, white noise, red noise, phase-randomized, IAAFT, and real-window nulls; match nuisance quantities as required.	Null generator
T4.6	Calibrate multiplicity	Use maximum-statistic or familywise/FDR control across trial Δ, filtrations, and summaries; avoid pointwise thresholds masquerading as detections.	Threshold tables
T4.7	Assess uncertainty	Evaluate bootstrap/Monte Carlo interval coverage and bias for each estimand; separate realization and pipeline-choice uncertainty.	Coverage maps
T4.8	Profile compute	Record runtime/memory by representation and sample size; set caching/chunking and ensemble sizes from precision targets.	Compute budget
T4.9	Freeze confirmation design	Select factors, repetitions, primary endpoints, practical-equivalence margins, and stopping rules before population validation.	Locked protocol v1

6.3. Minimum experiment cells

At minimum, include high/medium/low S/N; durations representative of one Kepler quarter, multiple quarters, one TESS sector, and multi-sector coverage; duty cycles near 0.5, 0.8, and 0.95; at least three mode lifetimes; correct and misspecified backgrounds; no gaps, random gaps, and mission windows; and ridge patterns with $\Delta\nu/2$ and $2\Delta\nu$ aliases. Replication counts are chosen from Monte Carlo standard-error targets, not convenience.

6.4. Deliverables and gate

- **D4.1** Analytic regression suite and derivation note.
- **D4.2** Validated stochastic solar-like oscillator simulator.
- **D4.3** Registered simulation/null design and immutable pilot archive.
- **D4.4** Detection, bias, coverage, and runtime maps.

Stage gate

G4 pass condition. Exact cases reproduce derived behavior; simulated inputs are recovered within prespecified tolerance; null false-positive rates and interval coverage are acceptably calibrated; and the planned confirmation experiment is computationally feasible. If a representation cannot be calibrated, it cannot support a detection claim and is either diagnostic-only or dropped.

7. Stage 5 - TDA core, representation audit, and statistical summaries

Stage objective

Implement tested topological primitives and compare five candidate representations without creating an uninterpretable feature collection.

Priority: Core. **Duration:** 5 weeks. **Depends on:** S4 analytic and null infrastructure.

7.1. Representations under review

Representation	Primary physical target	Required control / limitation
Light-curve filtration	Excursion hierarchy, asymmetry, activity/granulation morphology	Peak prominence and distributional summaries; not a primary p-mode comb estimator
Delay embedding	Periodicity, coherence, possible phase information, lifetime effects	PSD-preserving phase/IAAFT surrogates and Fourier/coherence controls
PSD superlevels	Peak hierarchy, envelope location, stable candidate selection	Ordinary peak prominence and smoothed-envelope baseline
Mode-frequency cloud	Comb/ridge geometry and degree separation	Nearest-neighbor/graph geometry without PH
Cylindrical echelle	Ridge connectivity and trial-spacing coherence	Classical autocorrelation, ridge linearity, image/graph controls

7.2. Tasks

ID	Task	Required work and evidence	Output
T5.1	Define filtrations	Specify domain, scalar function/distance, filtration direction, coefficients, maximum dimension, essential-class handling, and normalization for each representation.	Method spec
T5.2	Implement backends	Provide a common API for 1D exact persistence, Vietoris–Rips, cubical/periodic grids, and diagram distances/summaries.	TDA package

ID	Task	Required work and evidence	Output
T5.3	Cross-check backends	Compare independent libraries or exact solutions on small cases; define tolerance and deterministic ordering.	Regression tests
T5.4	Preserve localization	Map births/deaths to peaks, times, embedding points, pixels, radial orders, and ridges; expose selections in figures/tables.	Localization API
T5.5	Define compact summaries	Limit primary candidates to physically motivated totals, maxima, centroids, ridge lifetimes, stability distances, and calibrated p-values.	Feature registry
T5.6	Audit nuisance sensitivity	Test amplitude scaling, offset, monotone transformations, cadence, sample density, embedding dimension, smoothing, grid resolution, and periodic boundary handling.	Sensitivity report
T5.7	Quantify redundancy	Compute correlations, mutual/predictive information where justified, and conditional comparisons with classical controls across simulation development data.	Redundancy matrix
T5.8	Select/drop representations	Apply interpretability, calibration, incremental utility, stability, and compute criteria; record reasons without deleting failed work.	Selection memo

7.3. Deliverables and gate

- **D5.1** Versioned TDA core with analytic and cross-backend tests.
- **D5.2** Representation specification and parameter registry.
- **D5.3** Localization-aware diagnostic figures.
- **D5.4** Redundancy/ablation report and locked retained set.

Stage gate

G5 pass condition. Every retained statistic has a named physical target, a calibrated null, a simpler control, stable numerical behavior, and a prespecified role in a later claim. Statistics retained only because they look visually interesting fail the gate.

8. Stage 6 - Persistent topology of the power spectrum

Stage objective

Develop and validate the safest core method: persistence of background-normalized spectral peaks and physically localized estimators of the oscillation envelope and comb structure.

Priority: Core. **Duration:** 5–6 weeks. **Depends on:** S3–S5.

8.1. Tasks

ID	Task	Required work and evidence	Output
T6.1	Define normalized spectrum	Compare P/B, $\log(P/B)$, whitened residuals, and uncertainty-aware alternatives; freeze one primary and registered sensitivities.	Spectrum contract
T6.2	Compute exact H_0	Implement superlevel persistence with correct boundary/essential-class rules and link each class to birth frequency and merge saddle.	Peak diagram
T6.3	Verify prominence identity	Compare persistence lifetimes with an independently implemented peak-prominence control over analytic and realistic spectra.	Equivalence audit
T6.4	Estimate $\nu_{\max}$	Test persistence-weighted centroid, robust envelope location, and credible/confidence intervals; compare smoothing and pySYD.	$\nu_{\max}$ estimator
T6.5	Estimate comb structure	Construct persistence-weighted frequency-difference/autocorrelation scores; expose $\Delta\nu$, $\Delta\nu/2$, and $2\Delta\nu$ alternatives rather than forcing one.	Comb score
T6.6	Calibrate significance	Use maximum-statistic nulls over frequency and trial spacing; quantify false detection and false confidence.	Null calibration
T6.7	Stress-test background	Propagate background posterior/branches, smoothing, resolution, and window features into diagrams and estimators.	Robustness atlas
T6.8	Define quality flags	Create reason-coded flags for absent envelope, alias ambiguity, insufficient orders, window contamination, and unstable background.	QC schema

8.2. Primary comparisons

For $\nu_{\max}$, compare bias, median absolute error, interval coverage, and catastrophic-failure rate against smoothed-envelope and pySYD estimates. For comb recovery, compare with ACF/ACF2D, power-spectrum-of-power-spectrum, and known simulation truth. Report paired results by duration, S/N, duty cycle, and lifetime.

8.3. Deliverables and gate

- **D6.1** Localized PSD persistence pipeline.
- **D6.2** Topological $\nu_{\max}$ and comb estimator API with uncertainty.
- **D6.3** Null-calibrated detection and quality-flag tables.
- **D6.4** Topology-free equivalence/ablation analysis.

Stage gate

G6 pass condition. The method controls false alarms and interval coverage in locked simulations, handles spacing aliases explicitly, and provides either competitive robustness or a useful calibrated reliability flag. If PH is exactly equivalent to prominence without added inferential benefit, report the equivalence and retain it only as a stable organization/localization layer.

9. Stage 7 - Delay-embedding topology and phase-information test

Stage objective

Determine whether time-domain topology captures mode coherence, damping, or phase structure beyond information already contained in the PSD.

Priority: Publication/exploratory. **Duration:** 5 weeks. **Depends on:** S4–S5; may run parallel to S6.

9.1. Tasks

ID	Task	Required work and evidence	Output
T7.1	Select delay parameters	Compare physics-informed delays, autocorrelation minima, mutual information, and multiscale windows using simulation-only tuning.	Parameter policy
T7.2	Construct embeddings	Implement missing-data-aware sliding windows, normalization, dimensionality reduction used only for visualization, and sampling-density controls.	Embedding API
T7.3	Compute PH	Calculate H_0/H_1 and limited H_2 where supported; evaluate convergence with point subsampling and metric choice.	Diagram suite
T7.4	Calibrate coherence	Relate dominant H_1 persistence and diagram summaries to amplitude, linewidth, lifetime, quality factor, and mode mixture in simulations.	Response surfaces
T7.5	Apply PSD-preserving nulls	Compare original signals with phase-randomized and IAAFT surrogates using paired maximum-statistic tests.	Phase test
T7.6	Compare classical controls	Evaluate coherence time, autocorrelation decay, wavelet concentration, spectral entropy, and Fourier features under the same validation protocol.	Ablation table
T7.7	Test segmentation	Analyze quarter/window dependence, rolling/vineyard diagrams, and uncertainty due to finite realization.	Stability report

ID	Task	Required work and evidence	Output
T7.8	Make retain/drop decision	Retain only if it predicts a physical target or rejects PSD-preserving nulls with calibrated out-of-sample performance.	Decision memo

Stage gate

G7 pass condition. A positive path requires reproducible phase-sensitive information or useful lifetime/coherence calibration beyond classical controls. A null result is acceptable and publishable: it establishes that apparent delay topology is explained by PSD-preserving structure in the tested regime. In that case, freeze the negative result and remove delay PH from the critical path.

10. Stage 8 - Cylindrical topological echelle diagram

Stage objective

Formalize and validate the central innovation: a periodic echelle representation on $S^1 \times I$ whose persistent ridge structure scores trial large separations and stabilizes ridge geometry.

Priority: Core innovation. **Duration:** 8 weeks. **Depends on:** S3–S6.

10.1. Mathematical object

For trial spacing Δ, construct

$$E_\Delta(q, \theta) = \log\left[\frac{P\left(q\Delta + \Delta\theta/(2\pi)\right)}{B\left(q\Delta + \Delta\theta/(2\pi)\right)}\right], \qquad \theta \in S^1,$$

with periodic adjacency at the left/right echelle boundary. The stage compares cubical superlevel persistence, ridge graphs, and orderwise/zigzag persistence while retaining a simple ridge-coherence control.

10.2. Tasks

ID	Task	Required work and evidence	Output
T8.1	Define periodic complex	Specify discretization, radial-order interval, circular boundary cells, filtration, normalization, and treatment of incomplete orders.	Formal definition
T8.2	Verify boundary handling	Test ridges crossing the displayed edge and compare periodic versus incorrect rectangular adjacency on exact examples.	Boundary tests
T8.3	Construct ridge graph	Detect candidate maxima, connect adjacent orders with circular-distance and amplitude costs, and retain competing degree/ridge hypotheses.	Ridge graph API
T8.4	Define coherence score	Prespecify persistence/graph components of $\mathcal{T}(\Delta)$ and a topology-free control; normalize against number of orders and signal strength.	Score contract
T8.5	Scan trial spacing	Evaluate a physically motivated Δ grid with proper multiple-testing calibration and explicit half/double aliases.	Score curves

ID	Task	Required work and evidence	Output
T8.6	Quantify uncertainty	Bootstrap stochastic realizations/segments and propagate background/PSD branches to $\widehat{\Delta\nu}_{\mathrm{top}}$ and ridge locations.	Interval method
T8.7	Recover ridge angles	Estimate circular ridge centers and $\Delta\theta_{02}$, r_{02}, r_{01}, and r_{10} where mode visibility permits.	Ratio estimator
T8.8	Stress-test curvature	Inject glitches, ridge curvature, mode dropout, avoided crossings, rotation, and noise; document where the simple ridge model fails.	Failure atlas
T8.9	Compare baselines	Benchmark against ACF/PSPS, conventional echelle ridge fitting, Hough/linearity or graph-only methods, and published mode ratios.	Paired benchmark
T8.10	Prototype visualization	Produce a conventional/topological paired view with localized persistence features and uncertainty overlays suitable for papers and teaching.	Figure module

10.3. Deliverables and gate

- **D8.1** Formal periodic echelle complex and test suite.
- **D8.2** Open implementation of echelle field, ridge graph, persistence, and $\mathcal{T}(\Delta)$.
- **D8.3** Simulation-calibrated $\Delta\nu$ and ridge-angle estimators.
- **D8.4** Alias, failure-domain, and topology-free ablation report.
- **D8.5** Publication-quality topological echelle visualization.

Stage gate

G8 pass condition. Periodic-boundary tests pass; the score selects the correct alias at a frozen acceptable rate in detectable simulations; uncertainties have usable coverage; and performance or reliability is not explained entirely by a simpler ridge statistic. If utility is insufficient, retain the cylinder as a correct visualization/domain model and narrow claims to stability and geometry.

11. Stage 9 - Mathematical relations, bounds, and counterexamples

Stage objective

Separate exact identities, deterministic stability bounds, probabilistic calibration, and empirical regressions, and produce at least one nontrivial theorem-level contribution with physical interpretation.

Priority: Publication. **Duration:** 3–6 months, overlapping S6–S11. **Depends on:** analytic results and estimator definitions.

11.1. Target results

Target	Candidate statement	Required qualification
Peak identity	In one dimension, H_0 superlevel lifetime equals an appropriate peak-prominence difference.	Boundary and essential-component conventions explicit.
Delay ellipse	A sinusoid produces an ellipse with semiaxes determined by amplitude and $2\pi\nu\tau$.	Finite sampling, noise, and metric effects separated.
Diagram stability	$d_B(D(f), D(g)) \leq \|f - g\|_\infty$.	Link observational/processing perturbation to function-norm bound.
Ridge-angle relation	$r_{02} \simeq \Delta\theta_{02}/(2\pi)$.	Geometric relation; asymptotic and indexing assumptions explicit.
Spacing mismatch	Bound ridge drift/persistence loss as $\|\Delta - \Delta\nu\|$ increases over a finite order interval.	State mode curvature, missingness, and noise assumptions.
Stellar propagation	Convert bounded $\Delta\nu/\nu_{\max}$ error into first-order bounds on density, radius, mass, and g.	Include scaling-relation systematics separately.

11.2. Tasks

T9.1. State all objects, domains, coefficient fields, regularity assumptions, boundary conventions, and noise models before proofs.

T9.2. Prove exact idealized identities and encode each as a numerical/property-based regression test.

T9.3. Derive deterministic perturbation and ridge-drift bounds; identify constants that can be estimated from data or simulations.

T9.4. Develop probabilistic refinements for stochastic spectra when worst-case sup-norm bounds are too loose.

T9.5. Propagate estimator and topological perturbation bounds through seismic scaling relations while retaining independent astrophysical systematic terms.

T9.6. Search for counterexamples: merged peaks, edge births, missing radial orders, curved/crossing ridges, multimodal spacing scores, and stochastic realization changes.

T9.7. Classify each result as theorem, proposition, approximation, conjecture, or empirical relation; never mix categories in reporting.

T9.8. Obtain independent proof review and reproduce symbolic/numerical checks in a clean environment.

11.3. Deliverables and gate

- **D9.1** Formal definitions and theorem/proposition chapter.
- **D9.2** Proof-linked numerical test suite.
- **D9.3** Catalog of counterexamples and assumption violations.
- **D9.4** Uncertainty-propagation note linking topological/seismic and stellar quantities.

Stage gate

G9 pass condition. At least one nontrivial result beyond a direct citation is correct under explicit assumptions, independently reviewed, and informative for estimator design or interpretation. If new bounds are too loose, retain exact identities, counterexamples, and empirical calibration without overstating fundamentality.

12. Stage 10 - Rudy end-to-end benchmark and Paper I freeze

Stage objective

Produce a fully auditable, physically interpreted case study that connects classical analysis, topological constructions, null tests, uncertainty, and failure-aware conclusions for KIC 10963065.

Priority: Core. **Duration:** 5–6 weeks. **Depends on:** S3, S4, S6, and a usable S8 prototype; S7 optional.

12.1. Tasks

ID	Task	Required work and evidence	Output
T10.1	Freeze inputs	Lock Rudy product manifest, quality mask, registered preprocessing branches, reference table, code version, and configurations.	Analysis capsule
T10.2	Reproduce classical outputs	Regenerate PSD/background, window, $\nu_{\max}$, $\Delta\nu$, ordinary echelle, scaling parameters, and published-mode overlays.	Baseline figures
T10.3	Run spectral PH	Compute localized diagrams, envelope/comb statistics, null p-values, aliases, uncertainty, and quality flags.	Spectral results
T10.4	Run topological echelle	Scan trial spacing, show periodic ridge structure, estimate $\Delta\nu$ and ratios where supported, and compare graph-only/classical controls.	Echelle results
T10.5	Run delay test	If retained at G7, perform phase-preserving/null comparisons and lifetime interpretation; otherwise report the preregistered negative decision.	Delay result
T10.6	Analyze segment stability	Compare quarters, cumulative baselines, and bootstrap/stochastic variability; distinguish realization scatter from pipeline effects.	Stability table
T10.7	Degrade Kepler	Match TESS-like cadence, duration, noise, duty cycle, and real sector windows; locate actual TESS observations within detectability maps.	Transfer pilot

ID	Task	Required work and evidence	Output
T10.8	Conduct robustness branches	Run every registered detrending/background/mask/normalization branch; quantify conclusion stability rather than choosing the most favorable.	Multiverse report
T10.9	Freeze claims	Complete claim-evidence table, limitations, negative results, and reproducibility audit before manuscript drafting.	Claim ledger
T10.10	Draft Paper I	Write introduction, methods, simulations, Rudy results, ablations, discussion, and data/code availability; prepare archive.	Manuscript v1

12.2. Minimum figure set

The case study must include: observation/window overview; PSD plus background and mode envelope; conventional echelle; localized PSD persistence; trial-spacing topological score with aliases; cylindrical echelle/ridge graph; simulation placement of Rudy; sensitivity/uncertainty decomposition; surrogate result if delay topology is retained; and Kepler-to-TESS degradation.

12.3. Deliverables and gate

- **D10.1** Complete Rudy results bundle and machine-readable tables.
- **D10.2** One-command reproducibility capsule from public identifiers.
- **D10.3** Paper I manuscript and journal-ready figures.
- **D10.4** Independent reproduction and claim-evidence audit.

Stage gate

G10 pass condition. All headline outputs regenerate from a clean environment; conclusions survive registered branches or are narrowed; nulls and controls accompany every topological claim; and the case study makes no population-level generalization. This gate defines the minimum publishable and defense-ready scientific nucleus.

13. Stage 11 - Kepler LEGACY population validation

Stage objective

Test generalization, calibration, and valid operating domains on a star-grouped population with authoritative reference values and a one-time frozen holdout.

Priority: Publication; compact subset is core for defense. **Duration:** 3–5 months. **Depends on:** S10 method freeze.

13.1. Cohort strategy

Use three roles: development stars for debugging and interpretive examples; internal validation stars for model/threshold selection; and a frozen confirmation holdout opened once. Splits are by star, not light-curve segment. Stratification should cover $\nu_{\max}$, $\Delta\nu$, magnitude/S/N, mode linewidth, duty cycle, and evolutionary state as far as sample size permits.

13.2. Tasks

ID	Task	Required work and evidence	Output
T11.1	Reconcile cohort	Validate identifiers, products, literature values, uncertainty semantics, and exclusions for all 66 stars.	Cohort ledger
T11.2	Batch classical pipeline	Produce comparable PSD/background/window/global parameters/echelle products and quantify baseline failures.	Baseline catalog
T11.3	Batch topology	Run frozen representations and configurations; record failures without star-specific tuning.	Topology catalog
T11.4	Perform development QA	Repair general software defects only; log every code/configuration change and rerun affected development/internal validation results.	QA log
T11.5	Lock confirmation release	Freeze code, container, models, thresholds, exclusions, primary metrics, and statistical analysis before opening holdout.	Signed release
T11.6	Open holdout once	Run automated workflow without interactive adjustment; hash outputs and preserve the first valid run.	Confirmation archive

ID	Task	Required work and evidence	Output
T11.7	Estimate performance	Report paired bias, MAE/RMSE, coverage, false confidence, detection/failure, calibration, and practical-equivalence tests.	Performance tables
T11.8	Map operating domain	Stratify errors/failures by observational and stellar properties; use cautious partial pooling or descriptive intervals given sample size.	Failure atlas
T11.9	Validate ridge ratios	Compare topological/geometric r_{02} and related quantities with published individual-frequency estimates on adequate-S/N stars.	Ratio benchmark
T11.10	Audit leakage and multiplicity	Verify star grouping, tuning history, multiple comparisons, missingness, and any post-holdout deviations.	Audit report
T11.11	Prepare Paper II	Integrate formal construction, validation, counterexamples, and clear domain-of-validity statements.	Manuscript v1

13.3. Deliverables and gate

- **D11.1** Classical and topology-enhanced LEGACY benchmark catalog.
- **D11.2** Frozen holdout archive and leakage audit.
- **D11.3** Operating-domain/failure atlas with reason-coded quality flags.
- **D11.4** Paper II manuscript and machine-readable benchmark release.

Stage gate

G11 pass condition. Performance estimates are paired and uncertainty-aware; the final holdout was not used for tuning; failure domains are stated as explicitly as successful ones; and claims are supported across stars rather than by repeated segments. If global performance is heterogeneous, publish subgroup-specific validity instead of averaging away failures.

14. Stage 12 - Hybrid inference, quality control, and Kepler-to-TESS transfer

Stage objective

Test the strongest practical claim: whether topological information improves prediction, uncertainty calibration, or failure detection after conditioning on established asteroseismic features, and whether this value transfers to TESS-like observations.

Priority: Publication. **Duration:** 3–4 months. **Depends on:** S11.

14.1. Tasks

ID	Task	Required work and evidence	Output
T12.1	Define prediction tasks	Choose parameter regression, detectability, alias selection, uncertainty calibration, and catastrophic-failure prediction; define operational utility.	Task registry
T12.2	Build classical model	Fit a compact interpretable baseline using only conventional features with grouped nested validation.	Baseline model
T12.3	Build hybrid model	Add the locked small set of topology features; avoid high-dimensional persistence images unless sample size justifies them.	Hybrid model
T12.4	Test increment	Use paired out-of-sample differences, uncertainty intervals, calibration curves, decision curves, and practical-equivalence margins.	Increment report
T12.5	Create quality flags	Calibrate probabilities/reason codes for unreliable $\Delta\nu$, missing envelope, alias ambiguity, and domain shift.	Model card
T12.6	Design degradation experiment	Transform Kepler data to TESS-like cadence, exposure, duration, noise, duty cycle, contamination, and real sector windows with matched random seeds.	Degraded dataset
T12.7	Evaluate transfer	Compare classical and hybrid performance over degradation trajectories; distinguish resolution, noise, and window contributions.	Transfer curves

ID	Task	Required work and evidence	Output
T12.8	Analyze actual Rudy TESS	Apply the frozen method sectorwise and in justified combinations; compare with degradation predictions and report nondetections honestly.	TESS case study
T12.9	Recalibrate if authorized	If domain shift is systematic, fit a prespecified calibration layer on non-holdout transfer simulations; do not redesign features using test sectors.	Calibration note
T12.10	Decide broader TESS scope	Expand only if Rudy/degradation evidence supports a scientifically powered sample; otherwise stop at detectability/transfer limits.	Scope decision
T12.11	Prepare Paper III	Report practical increment or quantified redundancy/domain limits with model card and open transfer benchmark.	Manuscript v1

Stage gate

G12 pass condition. A positive complementarity claim requires locked out-of-sample improvement beyond a prespecified practical margin or materially better calibrated failure prediction. A positive transfer claim requires agreement between degradation predictions and actual observations within uncertainty. Otherwise publish a quantified redundancy or transfer-limit result.

15. Stage 13 - Physics-informed stellar models and advanced extensions

Stage objective

Use a deliberately bounded model grid to test structural sensitivity and determine whether topological ridge geometry responds to acoustic glitches, evolution, rotation, or mixed modes.

Priority: Extension. **Duration:** 3–6 months. **Depends on:** S8–S11; may be deferred post-defense.

15.1. Tasks and stop rules

T13.1. Define a compact design over mass, metallicity, age/evolutionary state, mixing length, and overshoot guided by the observed sample rather than an exhaustive stellar grid.

T13.2. Compute or reuse MESA structures and GYRE frequencies with recorded physics, convergence, and version metadata.

T13.3. Generate observable spectra using mode visibilities, linewidth/amplitude prescriptions, granulation, rotation/inclination where relevant, and mission windows.

T13.4. Test ridge curvature, acoustic-glitch signatures, separation ratios, and topology-score responses while conditioning on $\nu_{\max}$ and $\Delta\nu$.

T13.5. Compare matched models that share global seismic quantities but differ in internal structure; determine whether topology captures more than the known ratios.

T13.6. Explore subgiants/mixed modes only after the p-mode cylinder is validated; consider multi-parameter or zigzag approaches rather than forcing the solar-like main-sequence model.

T13.7. Stop the grid expansion if emulator/response uncertainty is dominated by mode-excitation prescriptions or if no conditional signal survives a prespecified pilot.

15.2. Deliverables and gate

- **D13.1** Reproducible compact stellar/model-frequency grid.
- **D13.2** Structural-response maps and matched-model tests.
- **D13.3** Extension decision for glitches, rotation, or mixed modes.
- **D13.4** Optional mathematical/methods paper or thesis chapter.

Stage gate

G13 pass condition. A retained extension must show a stable, interpretable response after conditioning on conventional seismic quantities and must fit within a clearly bounded manuscript question. Otherwise archive it as a future direction; it cannot delay Papers I–II or the thesis.

16. Stage 14 - Synthesis, publications, software release, and thesis

Stage objective

Consolidate the project into coherent papers, a defensible thesis, durable software/data products, and a transparent record of both positive and negative results.

Priority: Core. **Duration:** continuous writing; final intensive period 2–3 months. **Depends on:** completed core gates.

16.1. Tasks

ID	Task	Required work and evidence	Output
T14.1	Reconcile claims	Maintain one matrix linking thesis/paper claims to figures, tables, tests, assumptions, nulls, and limitations.	Evidence matrix
T14.2	Freeze releases	Tag code, lock environments, archive configurations, assign DOIs, and produce checksummed release manifests.	v1.0 release
T14.3	Curate data products	Publish allowed derived tables, figure inputs, benchmark catalogs, quality flags, metadata dictionaries, and regeneration instructions.	Data archive
T14.4	Complete manuscripts	Submit/revise Papers I–III according to attained evidence; do not wait for every extension before submitting the core.	Papers
T14.5	Assemble thesis	Integrate physical background, mathematical framework, methods, Rudy, population, transfer, negative results, and synthesis without duplicating inconsistent definitions.	Thesis draft
T14.6	Independent reproduction	Ask a collaborator to run priority workflows in a clean environment and record deviations, runtime, and undocumented assumptions.	Reproduction report
T14.7	Audit mathematics/citations	Independently check proof assumptions, equations, units, bibliography metadata, figure/table numbers, and claim-source alignment.	Audit checklist

ID	Task	Required work and evidence	Output
T14.8	Prepare defense	Build a narrative around questions and evidence; rehearse timing; prepare backup slides for nulls, aliases, leakage, uncertainty, and negative results.	Seminar package
T14.9	Archive research ledger	Preserve failed approaches, deviations, review decisions, compute notes, and future work so selective reporting is impossible.	Final ledger
T14.10	Plan continuation	Separate thesis-complete products from post-defense extensions and define ownership/maintenance for code and datasets.	Roadmap

Stage gate

G14 completion condition. The thesis claims are supported by frozen evidence; code and key outputs are independently reproducible; all departures from the registered plan are disclosed; negative results are integrated; and no optional extension is required to defend the central contribution.

17. Integrated deliverable register

Table 17.1.: Project deliverables, acceptance owner, and target stage.

ID	Deliverable	Acceptance evidence	Target
D0	Charter, protocol, repository	Clean smoke run; hypothesis/split audit	S0
D1	Evidence map and benchmark table	Verified primary citations; gap memo	S1
D2	Canonical data layer	Schema/provenance tests; frozen split	S2
D3	Rudy classical baseline	Pipeline agreement or reconciled differences	S3
D4	Simulator and null laboratory	Truth recovery, type-I error, coverage	S4
D5	Tested TDA core	Analytic/cross-backend tests; localization	S5
D6	Spectral topology estimators	Null calibration; classical ablations	S6
D7	Delay-phase result	PSD-preserving surrogate test	S7
D8	Topological echelle method	Periodic tests; alias/coverage benchmark	S8
D9	Mathematical results	Independent proof/test/counterexample review	S9
D10	Rudy case and Paper I	One-command regeneration; claim audit	S10
D11	LEGACY catalog and Paper II	Frozen holdout; leakage and failure audit	S11
D12	Hybrid/TESS result and Paper III	Out-of-sample increment/transfer evidence	S12
D13	Model-grid extension	Conditional structural response or stop memo	S13
D14	Thesis and archival release	Independent reproduction; DOI; defense	S14

17.1. Publication-ready definition

A manuscript is publication-ready only when its claim is supported by a locked result; methods run in a clean environment; figures derive from archived tables; uncertainty, nulls, and fair baselines are documented; limitations and negative controls are explicit; and data/code links target stable releases.

18. Quantitative decision framework

18.1. Primary metric families

Question	Primary metrics	Interpretation rule
Parameter recovery	Bias, MAE/RMSE, median fractional error, interval coverage	Paired by realization/star; report catastrophic failures separately
Detection	Sensitivity, specificity, precision, false-alarm rate, detection probability	Thresholds calibrated over complete scan, not pointwise
Reliability	Brier score, calibration error/curve, AUROC/AUPRC where appropriate	Prefer calibration and decision utility over ranking alone
Complementarity	Paired out-of-sample loss difference, likelihood/information increment, conditional tests	Require prespecified practical margin, not merely $p < 0.05$
Robustness	Sensitivity curves over duration, S/N, duty cycle, background, gaps	Identify operating boundary and uncertainty, not a single average
Topology stability	Bottleneck/Wasserstein distances, feature survival, localization change	Normalize physically and compare with known perturbation size
Transfer	Source/target error gap, calibration shift, detectability ratio	Attribute degradation by controlled interventions

18.2. Four-way gate decision

Pass. All core acceptance requirements met; proceed with frozen artifacts.

Pass with limitation. Evidence supports a narrower claim; limitation, affected regimes, and downstream restrictions are recorded.

Bounded repeat. A specific correctable defect is identified; repeat only named tasks with a deadline and unchanged holdout.

Stop/reframe. The method is redundant, uncalibrated, unstable, or infeasible; publish/record the boundary and move effort to the strongest surviving path.

18.3. Provisional practical thresholds

Exact thresholds are frozen after S4 pilots. Before then, planning uses: nominal coverage within Monte Carlo uncertainty of the target level; false-alarm control at the registered familywise/FDR level; no material increase in catastrophic failures; hybrid improvement exceeding both its uncertainty and a practical-equivalence margin; and acceptable alias identification conditional on detectability. Thresholds must be scientifically justified and cannot be selected to make a method pass.

19. Schedules and critical paths

19.1. Accelerated path to the May 2027 defense

Table 19.1.: Defense-critical schedule assuming launch in August 2026.

Month	Calendar	Primary work	Required freeze
M1	Aug 2026	S0 governance, S1 evidence map, S2 Rudy manifests and schemas	Charter, protocol, data identities
M2	Sep 2026	S3 Lightkurve/pySYD baseline, background/window/echelle audit	Classical reference and G3
M3	Oct 2026	S4 analytic cases, simulator, null families, initial coverage	Simulator/null design
M4	Nov 2026	S5 TDA core and S6 spectral persistence; ablations	Retained core statistics
M5	Dec 2026	S8 cylindrical echelle prototype; Rudy robustness branches	Topological method v0.9
M6	Jan 2027	S10 full Rudy synthesis; compact Kepler development subset	Rudy results and pipeline QC
M7	Feb 2027	Compact locked Kepler confirmation; S9 core proofs/bounds	Confirmation tables
M8	Mar 2027	Paper I submission; thesis chapter and figure freeze	Manuscript and evidence matrix
M9	Apr 2027	Thesis submission, independent reproduction, defense rehearsal	Thesis release candidate
M10	May 2027	Corrections, archive, seminar, defense buffer	Thesis/v1.0 defense release

Defense scope freeze: after February 2027, no new representation, large model grid, broad TESS sample, or high-dimensional learner enters the thesis core. New ideas go to the post-defense backlog unless they repair a correctness defect in an existing claim.

19.2. Full 30-month publication schedule

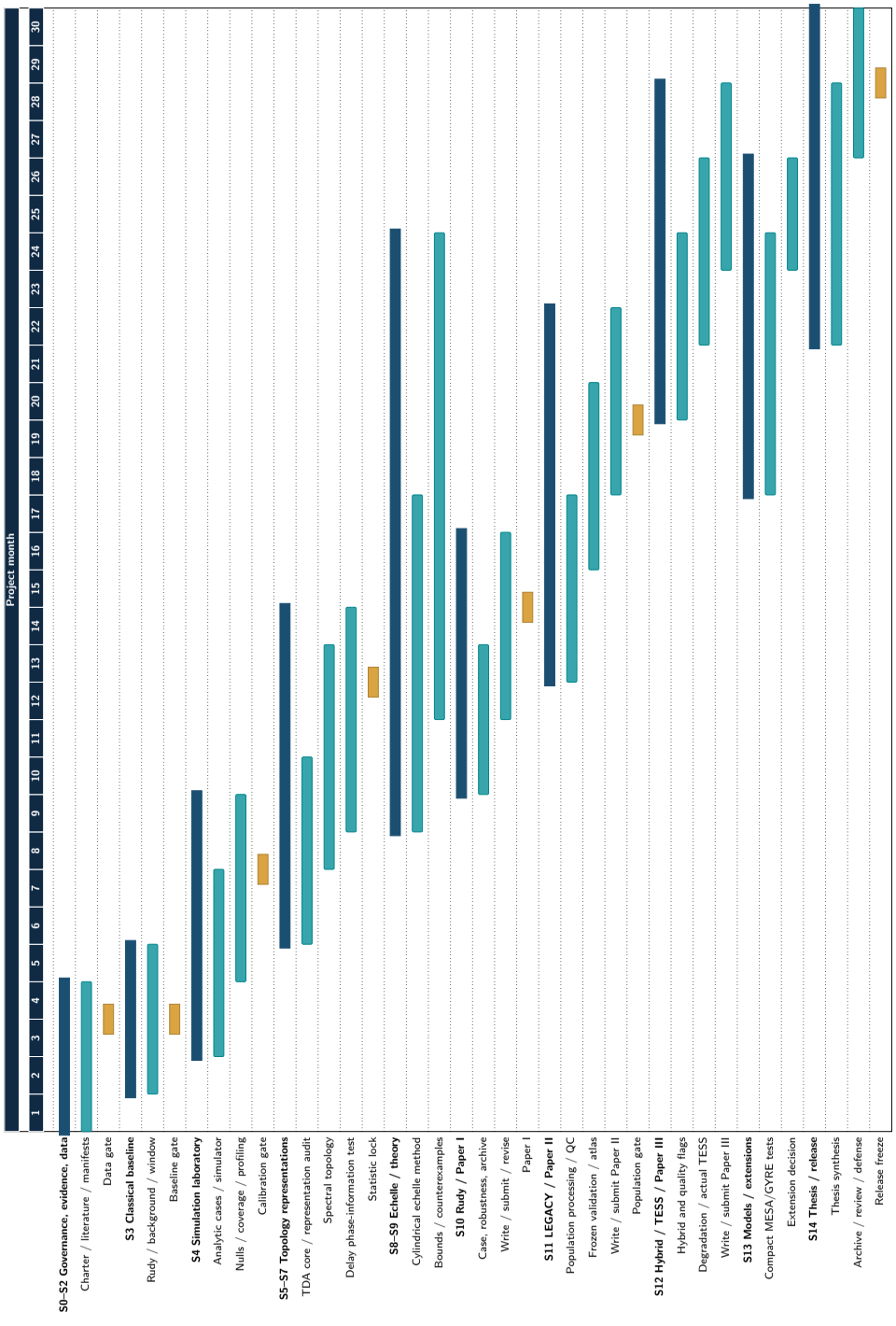


Figure 19.1.: Full publication roadmap. Writing, testing, and reproducibility continue across stages; stage bars indicate the main concentration of work.

19.3. Milestone review calendar

Month	Review	Evidence	Decision
M1	Launch	Charter, definitions, repository, access	Approve scope
M3	Classical gate	Rudy reproduction, QC, windows, references	Authorize topology
M5	Simulator review	Truth recovery, analytic tests, null validity	Freeze pilot factors
M7	Calibration gate	Type-I error, coverage, compute profile	Scale or simplify
M9	Representation review	Localization, redundancy, ablations	Retain/drop
M12	Statistic lock	Estimators, thresholds, Rudy blinded configuration	Authorize confirmation
M14	Paper I	Manuscript, capsule, independent review	Submit
M16	Population QC	Cohort ledger, exclusions, baseline failures	Freeze cohort
M19	Holdout gate	First-run archive, paired metrics, leakage audit	Set Paper II claims
M21–22	Paper II	Method, population, failure atlas, release	Submit
M24	Theory/extension gate	Proofs, utility, counterexamples, model pilot	Retain/defer extensions
M26	Transfer review	Hybrid increment, degradation, actual TESS	Set Paper III claim
M28	Release gate	Papers, package, data, thesis near-final	Corrections only
M30	Completion	Thesis/defense/archive/roadmap	Close project phase

Table 20.1.: Integrated risk register. P and I denote probability and impact: low (L), medium (M), high (H).

Risk	P	I	Early warning / mitigation	Fallback
Topology duplicates prominence	M	H	Near-perfect ablation equivalence in S5–S6; require localized/QC value	Publish exact equivalence and stability; remove novelty claim
Topological echelle reduces to ridge score	M	H	Graph-only control matches PH across pilots	Present correct cylindrical geometry and quantified redundancy
Delay topology has no phase information	H	M	Original and PSD-preserving surrogates indistinguishable	Freeze rigorous negative result; drop from critical path
Window artifacts mimic oscillations	M	H	High scores in background-only real-window simulations	Restrict domain; annotate aliases; require maximum-statistic null
Preprocessing dominates diagrams	M	H	Between-branch variation exceeds between-star/signal variation	Use topology as pipeline diagnostic; widen uncertainty; narrow claims
Rudy baseline mismatch	L–M	H	Classical values depend strongly on undocumented convention	Pause TDA and reconcile units/background/window/reference first
$\Delta\nu/2$ or $2\Delta\nu$ aliases dominate	M	M	Multimodal score curves in adequate-S/N pilots	Return candidate set/posterior and reliability flag, not forced estimate
LEGACY sample too small for flexible ML	H	H	Large validation variance or unstable selected features	Use compact interpretable models and paired statistics; avoid feature images
TESS Rudy detection inadequate	H	M	Degradation predicts low detection probability	Publish transfer/detectability limit; do not claim nondetection as method failure
Theoretical bound too loose	M	M	Worst-case bound many orders above empirical change	Seek local/probabilistic bound; retain exact cases and empirical calibration
MESA/GYRE expansion delays core	M	H	Grid design or convergence consumes more than one review cycle	Cap grid; defer S13; use published frequencies or asymptotic simulations

Risk	P	I	Early warning / mitigation	Fallback
Compute/storage pressure	L–M	M	Pilot runtime/storage exceeds allocation by factor two	Adaptive design, chunking, caching, fewer non-primary summaries
Defense deadline pressure	H	H	Core G6/G8/G10 slips beyond January 2027	Freeze extensions and full population; defend core plus compact validation

20.1. Change-control procedure

A proposed change records: motivation, affected hypothesis/stage/deliverable, whether data have been inspected, scientific benefit, leakage risk, schedule cost, alternative, decision, approver, and version. Changes after holdout access cannot be presented as confirmatory unless tested on a new independent dataset. Correctness fixes are mandatory but trigger a documented rerun of all affected outputs.

20.2. Schedule contingency

Maintain a two-week defense buffer and a one-month full-program buffer. If a core stage overruns by more than two weeks, first remove extension tasks, then reduce population breadth while preserving an independent validation, and only then narrow the central claim. Never save time by removing nulls, provenance, uncertainty, or baseline comparisons.

21. Recurring operational checklists

21.1. Weekly laboratory checklist

- Update issue board, experiment ledger, data/configuration hashes, and risk triggers.
- Confirm that each new result has a baseline, null where needed, and physical localization.
- Run unit, integration, analytic-regression, and smoke tests; record failures and environment.
- Regenerate at least one priority result from its immutable configuration.
- Convert completed analysis into manuscript-ready prose, table, or figure notes.
- Identify one decision for the next supervisory meeting.

21.2. Monthly supervisory review

1. What scientific question was answered this month, and with what evidence?
2. Which registered acceptance requirements passed or failed?
3. Did any null, control, or sensitivity analysis weaken the intended claim?
4. Which representations or extensions should be stopped?
5. What is the current critical-path delay and what scope is removed if it grows?
6. Are figures/tables reproducible and claims synchronized across paper and thesis drafts?
7. What single gate decision is needed before the next review?

21.3. Pre-holdout checklist

- Code tag, container/environment, input manifest, and output schema frozen.
- Cohort, exclusions, and missing-data rules frozen.
- Primary/secondary metrics and multiplicity correction frozen.
- Model/estimator hyperparameters and failure-return behavior frozen.
- Null thresholds, practical-equivalence margins, and uncertainty method frozen.
- Analysis script runs unattended on a mock holdout.
- One-time access authorization and deviation log prepared.

21.4. Pre-submission checklist

- Abstract conclusions match the registered estimands and actual evidence.
- No exploratory result is described as confirmatory.
- Every topological result has a non-topological control and appropriate null.
- Units, PSD normalization, radial-order convention, and periodic boundary are explicit.

- Tables contain uncertainty semantics, sample counts, exclusions, and quality flags.
- Figures regenerate from archived machine-readable inputs and remain legible in print.
- Mathematical statements have assumptions and correct theorem/approximation labels.
- References and dataset identifiers are verified; code/data links target stable releases.
- An independent clean-environment reproduction has been attempted and documented.

A. Stage report template

Every stage report should contain the following sections:

1. **Scope and version:** stage ID, dates, repository tag, environment, input manifest, responsible people.
2. **Questions and acceptance requirements:** copied from the frozen stage charter.
3. **Methods executed:** configurations, deviations, and reasons.
4. **Results:** machine-readable tables first, then figures and interpretation.
5. **Validation:** analytic tests, nulls, controls, uncertainty, coverage, and sensitivity branches.
6. **Negative results and defects:** failed hypotheses, software defects, unresolved anomalies, and impact.
7. **Deliverable register:** path/DOI, checksum, status, and acceptance evidence.
8. **Gate recommendation:** pass, pass with limitation, bounded repeat, or stop/reframe.
9. **Downstream consequences:** changed assumptions, removed tasks, new risks, and next-stage inputs.

B. Experiment ledger schema

Field	Required content
experiment_id	Immutable human-readable stage/task/run identifier.
hypothesis_id	Registered scientific hypothesis or diagnostic-only label.
status	Planned, running, completed, failed, superseded, or invalid.
data_manifest	Version/hash of raw and reference inputs.
split_role	Analytic, simulation pilot, development, validation, holdout, or descriptive case.
config_hash	Complete immutable analysis configuration hash.
code_version	Commit/tag and dirty-tree status.
environment	Lock/container ID and relevant hardware.
random_seeds	Explicit seed list or deterministic-seed rule.
estimand/metric	Quantity estimated and evaluation definition.
baseline/null	Exact comparison and null-generator versions.
outputs	Paths/IDs for tables, figures, logs, and serialized models/diagrams.
result	Short factual result without interpretation inflation.
decision	Retain, drop, repeat, reframe, or escalate; decision date/owner.
deviation	Difference from registered plan and whether data were inspected first.

C. Minimum Rudy benchmark checklist

1. Resolve KIC/TIC/HD/Gaia identifiers and coordinates.
2. Record exact Kepler quarters/cadence/products and TESS sectors/products.
3. Reproduce the documented Lightkurve analysis with versioned dependencies.
4. Run an independent classical pipeline and reconcile definitions.
5. Validate time units, frequency grid, Nyquist frequency, PSD normalization, and background units.
6. Compute window functions and mark aliases.
7. Produce ordinary echelle with published mode frequencies.
8. Execute all registered preprocessing/background branches.
9. Compute localized PSD persistence and calibrated null distribution.
10. Compute trial-spacing curve with half/double aliases visible.
11. Construct periodic cylindrical echelle and compare rectangular-boundary error.
12. Estimate ridge positions/ratios only where visibility and uncertainty support them.
13. Run PSD-preserving surrogates if delay topology remains active.
14. Perform quarter/cumulative-baseline stability analysis.
15. Degrade Kepler to TESS-like observing conditions and compare with actual sectors.
16. Regenerate every headline output from a clean environment and public IDs.

D. Minimum publishable outcomes under failure

Failed intended claim	Scientifically defensible result	Required evidence
PH improves PSD peak analysis	PH is exactly/approximately prominence with formal stability but no accuracy gain	Proof/controlled equivalence, nulls, robustness comparison
Delay PH adds phase information	Delay topology is explained by the PSD in solar-like stochastic regimes	PSD-preserving paired surrogates over simulations and Rudy
Topo-echelle improves $\Delta\nu$	Correct cylinder/periodic representation adds visualization or QC but not point accuracy	Alias/failure atlas and fair graph/classical controls
Ridge topology recovers ratios broadly	Recovery is valid only in high-S/N, multi-order main-sequence cases	Calibrated operating-domain map and explicit exclusions
Hybrid model improves parameters	Topology predicts classical-estimator failure or uncertainty miscalibration	Locked out-of-sample reliability and decision analysis
Rudy TESS oscillations are recovered	TESS configuration yields a justified nondetection/detectability upper limit	Kepler degradation, window-aware simulations, sectorwise analysis
New bounds are tight	Exact idealized results and counterexamples clarify why general bounds are loose	Correct proofs, numerical verification, physical interpretation

E. Publication and thesis mapping

Product	Core stages	Minimum central claim	Indicative venue/status
Paper I	S3–S6, S10	Null-calibrated spectral topology and auditable Rudy benchmark	MNRAS, A&A, or ApJ
Paper II	S8–S11, S9	Cylindrical echelle formulation plus locked Kepler population validation	A&A, MNRAS, or ApJ
Paper III	S12	Conditional utility/redundancy and Kepler-to-TESS transfer	AJ, ApJS, or MNRAS
Optional methods paper	S7/S9/S13	Phase-information limit, bounds, ridge geometry, or model response	Applied mathematics/methods venue
Thesis core	S0–S10 plus compact S11	Defined, calibrated framework; Rudy; independent Kepler test; honest limitations	May 2027 target
Full research program	S0–S14	Population, theory, mission transfer, releases, and post-defense extensions	30-month horizon

F. Closing execution principle

The project succeeds by answering a small number of difficult questions convincingly, not by maximizing the number of diagrams or algorithms. Every stage therefore forces a choice: retain a construction because it has a physically interpretable and validated role, narrow it to quality control or visualization, or stop it and preserve the negative evidence. This gate-based structure protects the doctoral deadline while retaining a clear path to the more ambitious population, mathematical, and cross-mission publications.

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